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Unit 3 : Session 9 : SLO 1 — Restoring Division Algorithm (Procedure)

■ Section A: Match the Following (Remember)

- A. Sign Bit → 4. Indicates positive or negative
- B. Negative Remainder → 5. Triggered when subtraction fails
- C. Quotient Register → 3. Stores result bits (0 or 1)
- D. Shift Step → 2. Logical left shift on registers
- E. Restoring Step → 1. Add divisor to restore remainder

■ Section B: Fill in the Blanks (Understand)

- The restoring division algorithm is primarily used for BINARY division of binary numbers.
- If subtraction in a cycle produces a NEGATIVE remainder, the algorithm restores the previous value.
- The REMAINDER register is shifted left with the quotient during each iteration.
- The divisor is SUBTRACTED from the remainder in each cycle.
- The quotient bit is set to 1 if the subtraction is successful.